Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

Roll Number: 2520030053
Name: V. Karthik

Cloud Infrastructure and Services

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LAB RECORD

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Skill Experiment - 1

AIM: To create an IAM user with programmatic access, generate and download Access Key ID and Secret Access Key credentials, and configure AWS Free Tier usage notifications and a monthly billing budget with email alerts.

Description:

This experiment demonstrates basic AWS identity management and cost monitoring. An IAM user is created for programmatic access only, without enabling console sign-in. An access key pair (Access Key ID and Secret Access Key) is generated and downloaded for programmatic interaction with AWS services. The experiment also enables AWS Free Tier usage notifications through Billing Preferences and creates a monthly cost budget so that email alerts can be sent when the configured spending threshold is reached. The supplied output screenshots show the created zero-spend budget, its healthy status, zero recorded spend at the time of capture, and the email alert recipient with verification still pending.

Steps:

Part A: Create an IAM User with Programmatic Access Only

1. Sign in to the AWS Management Console using the root account or an account with administrative permissions.
2. Search for IAM in the AWS Console search bar and open Identity and Access Management (IAM).
3. From the navigation pane, select Users and click Create user.
4. Enter a suitable username. Do not enable AWS Management Console access because the user is intended for programmatic access only.
5. Click Next. Under Set permissions, choose Attach policies directly and select the policy required for the practical (the supplied experiment sheet gives Administrative access as an example).
6. Review the configuration, click Create user, and verify that the new user appears in the Users list.

Part B: Generate and Download Access Keys

7. Open the newly created IAM user and select the Security credentials tab.
8. Scroll to Access keys and click Create access key.
9. Select the appropriate use case, such as Command Line Interface (CLI), acknowledge the recommendation, and click Next.
10. Optionally add a description tag, then click Create access key.
11. Note the Access Key ID and Secret Access Key. Download the credentials as a CSV file and store it securely. The secret access key should not be shared or exposed.
12. Click Done and confirm that the access key is listed for the IAM user.

Part C: Configure AWS Free Tier Usage Alerts

13. Open Billing and Cost Management using the AWS Console search bar.

14. Open Billing preferences and locate the Alert preferences section.

15. Enable Receive AWS Free Tier usage alerts. Enter an additional email address if required and update the settings.

16. Verify that the Free Tier usage alert option is enabled.

Part D: Create a Monthly AWS Budget and Configure Email Alerts

17. In Billing and Cost Management, open Budgets and click Create budget.

18. Choose the Zero spend budget template, or choose Customize (advanced) to configure a Cost budget manually.

19. For a custom budget, set the budget period to Monthly, choose a recurring budget, set the start month, select a fixed budgeting method, and enter the budget amount and scope.

20. Add an alert threshold. Set the threshold percentage or amount and choose whether the alert is triggered by actual or forecasted spend.

21. Enter the email recipient for budget notifications. Skip optional budget actions if they are not required for the experiment.

22. Review the budget configuration and click Create budget.

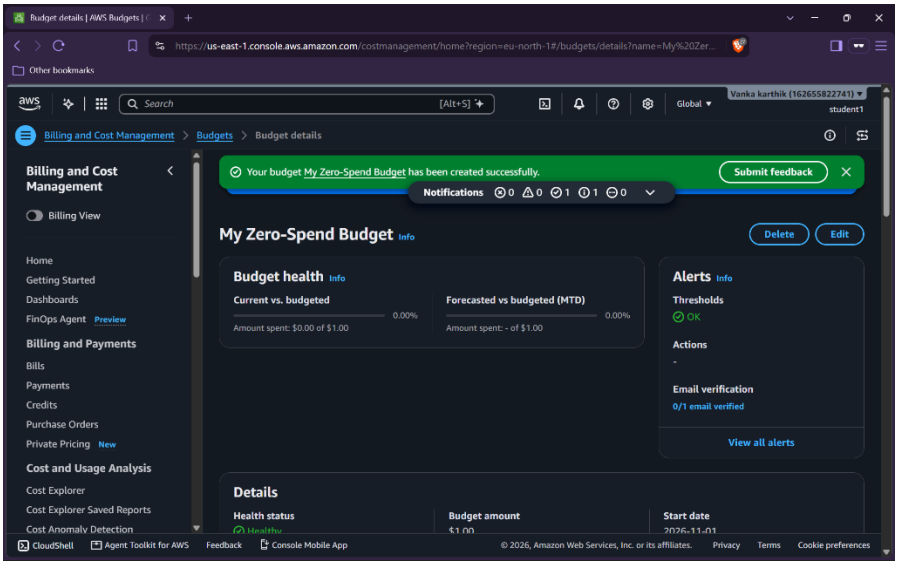
23. Verify that the budget appears in the AWS Budgets dashboard. Check the recipient's verification status and complete email verification if AWS requests it.

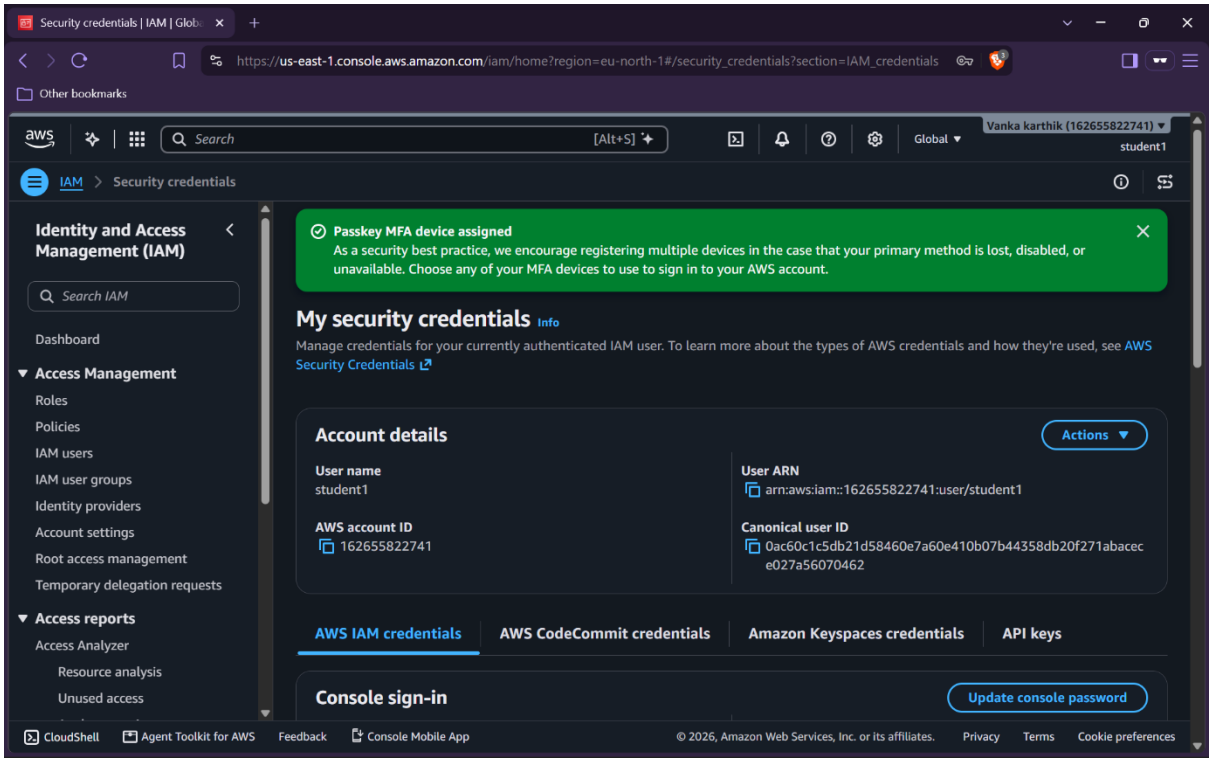
Output:

The AWS Budgets page displays a budget named “My Zero-Spend Budget”. Its health status is Healthy, the budget amount is $1.00 with a Monthly period, and the current spend shown is $0.00 of $1.00 (0.00%). The Alerts panel shows the threshold status as OK. The email verification section lists one recipient and shows its verification status as Pending; therefore, the screenshots confirm that the budget and alert recipient were configured, but do not show that email verification was completed.

Output Screenshot 1: AWS Zero-Spend Budget details and budget health.

Output Screenshot 2: Budget details and email alert recipient verification status.





Result: The monthly zero-spend budget was created successfully and its dashboard status was Healthy. The configured email recipient was still Pending verification in the captured output.